

Syllabus
Academic Year 2024-2025

1. General Information	
Course title	Computational Mathematics
Degree cycle (level)/major	6B06103 Big Data Analysis
Year, term	2, 3
Number of credits	5
Language of delivery:	English
Prerequisites	Calculus 1, Calculus 2, Linear algebra, Introduction to Programming, Object-Oriented Programming
Postrequisites	Research Project / Capstone Project
Lecturer(s)	1. Bibinur Meirbekova, PhD, Assistant Professor, b.meirbekova@astanait.edu.kz , Astana IT University, Expo, C1 block, office C1.1.337
2. Goals, Objectives and Learning Outcomes of the Course	
1. Course Description	This course is an advanced introduction to numerical linear algebra and related numerical methods. Topics include direct and iterative methods for linear systems, eigenvalue decompositions and LU factorizations, stability and accuracy of numerical algorithms. Numerical differentiation and integration. Interpolation. Least squares and regression analysis. Numerical solution of ordinary differential equations. Introduction to error analysis.
2. Course Goal(s)	The goal of the course is to equip students with the mathematical knowledge and computational skills necessary to model, analyze, and solve complex real-world problems. By integrating theoretical concepts with practical applications, the course aims to foster critical thinking, enhance problem-solving abilities, and prepare students for advanced studies or careers in fields such as engineering, data science, finance, and scientific research. Through hands-on experience with numerical methods, programming, and mathematical modeling, students will develop a robust understanding of how to leverage computational tools to derive meaningful insights from data and mathematical formulations.
3. Course Objectives:	To enhance the problem-solving skills of engineering students using an extremely powerful problem-solving tool namely numerical method. The tool is capable of handling large system of equations, non-linearities and complicated geometries that are not uncommon in engineering practice and that are often impossible to solve analytically
4. Skills & Competences	• Proficiency in developing numerical algorithms; • Implementing mathematical models through programming languages; • Analyzing computational errors;

	• Solving linear and nonlinear equations; • Performing advanced numerical integration; • Creating interpolation techniques; • Designing root-finding strategies; • Applying optimization methods; • Understanding computational complexity; • Critically evaluating mathematical problem-solving approaches across scientific and engineering domains.
5. Course Learning Outcomes:	By the end of this course students will be able to: • Understand numerical techniques to find the roots of non-linear equations and solution of system of linear equations. • Use Taylor Series to approximate functions and evaluate the approximations error. • Program algorithms to locate the roots of nonlinear equations. • Program algorithms to solve linear system of equations. • Smooth engineering collected data using least square method. • Understand the difference operators and the use of interpolation. • Understand numerical differentiation and integration and numerical solutions of ordinary differential equations.
6. Methods of Assessment	The expected learning outcomes for the course will be assessed through graded activities and ungraded activities. The graded activities include exams, homework, quizzes, and midterms. The ungraded activities will be used to monitor your progress. A variety of these ungraded assessment techniques may be employed, including problems to be completed during class, direct questioning of students, answering students' questions in class, one-minute classroom assessment techniques, and discussions during office hours.
7. Reading List	Assigned reading materials and presentations should be read prior to class. Class lectures and discussions will proceed with supplemental and advanced topics, which could be difficult to understand unless students have read the assigned material. Readings are listed in the schedule section. All necessary updates and / or changes to the course will be reflected in the Learning Management System (moodle.astanait.edu.kz). <u>Basic Literature:</u> 1. Lecture notes (available at learn.astanait.edu.kz, moodle.astanait.edu.kz) 2. B.S. Grewal, "Numerical Methods in Engineering & Science", Khanna Publication, Ed. 9th., 2014 <u>Supplementary literature:</u> 1. E. Kreyszing "Advanced Engineering Mathematics" John Wiley & Sons, Inc, Ed 10th, 2011
1. Resources	Online journals, article, papers, books and internet resources as well as online emulators and online software for simulation.
2. Course policy	Course and University policies include: Attendance: Attendance is not allocated any grading points in the marking scheme, but is compulsory to pass the course. Normally students are required to

	achieve course attendance of minimum 70% to get admitted to the examination rubric. In case a student misses 30% or more class sessions without a valid excuse the instructor has the right to mark him as “not graded”. In such case a student is not admitted to the exam and automatically fails the course. In cases, when a student misses class session due to valid reasons (is excused by the instructor or the dean’s office) he or she has to confirm the absence reason using a valid document in accordance with the academic policy of AITU. It should be NOTED that in cases when a student is excused for 30% of the scheduled class sessions or more he or she has to study material provided under the course on their own. Course instructor might provide additional opportunities to submit missed graded pieces of work during office hours or conduct alternative assessment exercises using method of his or her choosing. Preparation for Class: Class participation is a very important part of the learning process in this course. Although not explicitly grade, students will be evaluated on the QUALITY of their contributions and insights. Quality comments possess one or more of the following properties: - Offers a different and unique, but relevant, perspective; - Contributes to moving the discussion and analysis forward; - Builds on other comments. Class work: The duration of each lecture and practical lesson is 50 minutes for offline class, and 40 minutes for online class. Students are expected to complete all readings and assignments ahead of time, attend class regularly and participate in class discussions. In case of systemic student’s misconduct, the student can be dispensed from the classes. Being late on class: When students come to class late, it can disrupt the flow of a lecture or discussion, distract other students, impede learning, and generally erode class morale. Moreover, if left unchecked, lateness can become chronic and spread throughout the class. Therefore, the being late to the class is not welcome and can have restriction activities by the course instructor. Attestation I and II: Students, who score less than 25% for Attestation period I or Attestation period II (RK1/RK2) automatically fail the course. Home work / Assignments: The assignments are designed to acquaint students with the theoretical knowledge and practical skills required for the course. The textbook readings will be supplemented with materials collected from recent professional articles and journals. In case of using someone’s work (papers, articles, any publications), all works must be
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	properly cited. Failure to cite work will be resulted as a cheating from the students and may be a subject of additional disciplinary measures. Late submissions: Most assignments will be discussed in class on the due date. It is expected that all work will be submitted on time. All gradings are based using a percentage grading scale. <i>In the case of some extraordinary event, students should notify the course instructor and request an extension of the deadline for submission. If approved, a new date will be given to the student depending upon the circumstances by the instructor.</i> Final exam: The final exam for the course “Computational Mathematics” includes eight theoretical and practical tasks for 100 minutes. Students will be given tasks in written form and must give their answers in written forms. Laptops and mobile devices can only be used for classroom purposes when directed by the course instructor. Misuse of laptops or handheld devices will be considered a breach of discipline and appropriate action will be initiated by the instructor. Online lessons can be used in case if there won’t be a chance to make offline traditional lessons. It must not discourage the interest and enthusiasm of students. The main software to run the online lessons is Microsoft Teams for video calls and live webinars, and Moodle (moodle.astanait.edu.kz) as a Learning Management System. Also, some alternatives such as Telegram, Zoom, or other messenger may be involved as an additional workaround. Cheating and plagiarism are defined in the Academic conduct policies of the university and include: 1. Submitting work that is not your own papers, assignments, or exams; 2. Copying ideas, words, or graphics from a published or unpublished source without appropriate citation; 3. Submitting or using falsified data; 4. Submitting the same work for credit in two courses without prior consent of both instructors. Any student who is found cheating or plagiarizing on any work for this course will receive 0 (zero) for that work and further actions will also be taken regarding academic conduct policies of the university. Academic Conduct Policies of the university: The full texts of all the academic conduct code will be posted to the students using Learning Management System (moodle.astanait.edu.kz). Contacting the Course instructor: The easiest and most reliable way to get in touch with the course instructor is by email. Students must feel free to send email if you have a question related to the course. Instructor responds as soon as they can but not always instantaneously. Besides that, students are also welcomed to arrange a one-to-one meeting with the instructor by their office during office hours to discuss the class
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	using both offline and online.
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* Final exam can be given by instructor in the any other forms such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.

3. Course Content

#	Abbreviation	Meaning
1	ISIS	Instructor-supervised independent work
2	SIS	Students' independent work
3	IP	Individual project
4	PA	Practical assignment
5	LW	Laboratory work
6	MCQ	Multiple choice quiz

3.1 Lecture, Practical/Seminar/Laboratory Session Plans

Week No	Course Topic	Lectures (H/W)	Practice sessions (H/W)	Lab. sessions (H/W)	TSIS (H/W)	SIS (H/W)
1-2	Approximations and Errors in Computation. Introduction, Significant digits, Rounding error, Truncation error, Absolute and relative error, Error propagation. Solution of Algebraic and Transcendental Equations. Roots of Non-Linear Equations: Graphical method, Bisection method, False position Method, Secant method, Iteration method, Newton-Raphson Method, Muller's method for complex and multiple roots.	6	4	0	2	18
3-4	Solution of Simultaneous Algebraic Equations. Jacobi's iteration method, Gauss Elimination method by pivoting, Gauss – Jordan method, Gauss – Seidel method. Relaxation method, Ill-Conditioned Equations, Convergence of iteration methods. Matrix Inversion and Eigenvalue Problem. Iterative Method, LU factorization Power method, Jacobi's method, Given's Method, House-Holder's Method.	6	4	0	2	18

5	Empirical Laws and Curve-Fitting. Graphical Method, Method of Least Squares, Fitting curves. Method of Group Averages, Laws Containing Three Constants.	3	2	0	1	9
6-7	Finite Differences. Difference Operators: Forward and Backward difference operators and table. Interpolation. Newton's Forward/Backward Interpolation Formula. Central difference Interpolation Formula. Lagrange's Interpolation Formula. Divided differences, Newton's Divided Difference Formula Spline Interpolation.	6	4	0	2	18
8-9	Numerical Differentiation. Derivatives using Newton's forward/ backward difference formula, Derivatives using unequally spaced values of argument, Maxima and Minima of a Tabulated Function. Numerical Integration. Newton – Cotes integration formula, Trapezoidal rule, Simpson's rule, Boole's rule and Weddle's rule, Romberg's Integration. Errors in Quadrature Formulae.	6	4	0	2	18
10	Numerical Solution of Ordinary Differential Equations. Picard's Method, Taylor series method, Euler and modified Euler method, Runge-Kutta methods.	3	2	0	1	9
	Total hours: 150	30	20	0	10	90

3.2 List of Assignments for Student Independent Study

№	Assignments (topics) for Independent study	Hours	Recommended literature and other sources (links)	Form of submission
1	2	3	4	5
1	Advanced Error Analysis in Numerical Methods.	18	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
2	Nonlinear Dynamics and Chaos Theory. False Positions Method, Muller's method	18	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
3	Adaptive Numerical Methods for	9	Reading list, books,	Reading list,

	Differential Equations		internet resources Electronic version	books, internet resources Electronic version
4	Multivariable Interpolation Techniques	18	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
5	Numerical Methods for Partial Differential Equations	18	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
6	High-Dimensional Numerical Integration	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version

4. Student Performance Evaluation System for the Course

Period	Assignments	Number of points	Total
1 st attestation	Assignments: assignment 1, assignment 2, assignment 3 Mid Term Quiz	60 20 20 20 40 40	100
2 nd attestation	Assignments: assignment 4, assignment 5, assignment 6 End Term Quiz	60 20 20 20 40 40	100
Final exam	Quiz		100
Total	0,3 * 1st Att + 0,3 * 2nd Att + 0,4*Final		100

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

Letter Grade	Numerical equivalent	Percentage	Grade according to the traditional system
A	4,0	95-100	Excellent
A-	3,67	90-94	
B+	3,33	85-89	
B	3,0	80-84	Good
B-	2,67	75-79	
C+	2,33	70-74	
C	2,0	65-69	Satisfactory
C-	1,67	60-64	
D+	1,33	55-59	

D	1,0	50-54	Fail
FX	0	25-49	
F	0	0-24	

Based on the specific grade for each assignment, and the final grade, following criteria must be satisfied:

Grade	Criteria to be satisfied
90-100	- Work would be worthy of further dissemination under appropriate conditions - Mastery of advanced methods and techniques at a level beyond that explicitly taught - Ability to synthesize and employ in an original way idea from across the subject - Outstanding command of critical analysis and judgment
80-89	- Excellent range and depth of attainment of intended outcomes - Mastery of a wide range of methods and techniques - Evidence of study and originality of what has been taught - Able to display a command of critical analysis and judgement
70-79	- Attained all the intended learning outcomes for a unit - Able to use well a range of methods and techniques to come to conclusions - Able to employ critical analysis and judgement
60-69	- Some limitations in attainment of learning objectives, but has managed to grasp most of them - Able to use most of the methods and techniques taught - Evidence of study and comprehension of what has been taught but grasp insecure - Some grasp of the issues and concepts underlying the techniques and material taught, but weak and incomplete
50-59	- Attainment of only a minority of the learning outcomes - Able to demonstrate a clear but limited use of some of the basic methods and techniques taught - Weak and incomplete grasp of what has been taught - Deficient understanding of the issues and concepts underlying the techniques and material taught
25-49	- Attainment of nearly all the intended learning outcomes deficient - Lack of ability to use at all or the right methods and techniques taught - Inadequately and incoherently presented - Wholly deficient grasp of what has been taught - Lack of understanding of the issues and concepts underlying the techniques and material taught
0-24	No significant assessable material, absent or assessment missing a must pass component

5. Methodological Guidelines

Assessment is administered continuously throughout the course. The students are rated against their performance in continuous rating administered throughout the semester (60%) and summative rating done during the examination session (40%), total 100%. Continuous rating is students' on-going performance in class and independent work. Class work is assessed for attendance, laboratory works' defense and in- class assessments.

- **ISIS (Instructor Supervised Student Independent Study)** -comprises presentation to be done by students independently and checked by instructor.

- **Mid-term and End-term** is a review of the topics covered and assessment of each student's knowledge. The form of the midterm and end term exams is complex.
- **Final assessment** is a combination of both individual (team) project (report) and oral presentation (slides) to evaluate the students' academic performance and professional skills. At the completion of this course each student has to submit an online project version conforming to the project outline, as well as to prepare and present a slide presentation that follows the presentation outline. Project iterations would be required to submit as well. Students should submit the written Report of Final project and slide-Presentation 3 days before the Final exam day.

6. Lecturer (lecturers) approvals Full name Job title Date Sign

Full name	Job title	Date	Sign
Meirbekova Bibinur	Assistant Professor		

**Director of Department of
Computational and Data Sciences**

M.Zh. Sergaziyev